

Yathin Reddy Duvuru

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EDUCATION

New York University, Tandon School of Engineering <i>M.S. in Computer Science (GPA: 3.95/4.0)</i>	New York, NY Sep 2025 – May 2027
SRM Institute of Science and Technology <i>B.Tech. in Computer Science and Engineering (GPA: 9.49/10.0)</i>	Chennai, India Sep 2021 – May 2025

TECHNICAL SKILLS

ML: PyTorch, TensorFlow, Scikit-learn, NumPy, SeaBorn, XGBoost, MLFlow, SHAP, Captum, HF Accelerate
Generative AI: LangChain, LangGraph, LlamaIndex, Nano-GraphRAG, Qdrant, ChromaDB, Ollama
Languages & Backend: Python, C/C++, SQL, JavaScript, FastAPI, Flask, PostgreSQL, MongoDB
Cloud & DevOps: AWS (API Gateway, Lambda, DynamoDB, DAX), Docker, Kubernetes, Terraform, Git

EXPERIENCE

New York University, Center for Urban Science and Progress <i>TREx Research Fellow</i>	Brooklyn, NY June 2026 – Present
- Created a comparative evaluation pipeline for a fine-tuned RemoteCLIP model (RoofNet), applying XAI frameworks to audit biases and interpret features across 530+ satellite images. - Migrated the GeoFlood Studio platform from Cesium Ion to a self-hosted TiTiler/Nginx architecture on NYU on-prem infrastructure, cutting cloud dependencies and increasing storage capacity 10x.	
New York University, Stern School of Business <i>Graduate Assistant Developer</i>	Brooklyn, NY Mar 2026 – Present
- Architected migration of the CI Catalyzer platform to a Multi-Agent LangGraph system, boosting task completion rates by 35% through autonomous orchestration and fault isolation. - Designed a high-concurrency, tiered rate-limiting system via AWS API Gateway, Lambda, and Amazon DAX, achieving sub-10ms lookup latency with DynamoDB fallback.	
SRM Institute of Science and Technology <i>Undergraduate Researcher</i>	Chennai, India Aug 2024 – May 2025
- Researched LSTM-based time-series forecasting for solar power with XAI integration, achieving a 40% reduction in training time and 10% RMSE reduction over statistical baselines. - Awarded the Department Silver Medal at SRM Research Day 2025; published in Springer LNNS.	

PROJECTS

Full-Stack RAG Document QA System <i>FastAPI, React, Qdrant, Redis, Celery, PostgreSQL</i>	Jun 2026
- Engineered a hybrid RAG pipeline using Reciprocal Rank Fusion (RRF) to fuse BM25 and semantic search, serving top-k context chunks with SSE answer streaming. - Built a Redis HNSW semantic cache with a two-pass hydration strategy, cutting LLM API costs and delivering sub-75ms response latency on repeated queries.	
Automated Recipe Normalization Engine <i>Triton, Kubernetes, Ansible, MLflow, HF Accelerate</i>	May 2026
- Fine-tuned T5 variants on a 150K recipe dataset, achieving 0.70 ROUGE-L; accelerated multi-GPU training ≈2x via FP8 precision over BF16 with Hugging Face Accelerate. - Provisioned a 5-node Kubernetes cluster via Ansible playbooks, cutting setup time by 90%; deployed to Triton Inference Server with sub-3s latency monitored via Prometheus/Grafana.	
GraphRAG-Based AI Tutor <i>Nano-GraphRAG, Qdrant, Chainlit, Ollama, Docker</i>	Dec 2025
- Deployed a GraphRAG system fusing vector similarity search with knowledge graph traversal (Nano-GraphRAG), enabling multi-hop reasoning and improving answer relevance by 30%. - Developed an automated graph construction workflow using Qwen3-32B, reducing manual schema design effort by 80% while processing 255+ documents for downstream retrieval.	

PUBLICATIONS

Yathin Reddy D., Seshank M., Saranya P. *Photovoltaic Cell Power Forecasting Using LSTM with XAI Integration*. Springer LNNS, 2024. [doi:10.1007/978-981-96-7514-2_22](https://doi.org/10.1007/978-981-96-7514-2_22)